

Curating a Virtual Exhibition through Generative AI and VR Sketches

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Figure 1: The Space planning and system design of *CanalWalk*: (a-e) exhibition halls and galleries; (d-g) interaction mechanisms include treasure hunt, photo-taking and graffiti creation; (h) VR sketches using OpenBrush in design process.

ABSTRACT

This paper introduces an innovative technical form to enhance visiting experiences that respect the user perspectives about cultural heritage and the integrity of cultural contents. We curate an interactive art exhibition, *CanalWalk*, using an autobiographical design approach to reconstruct cultural heritage through co-creation. Our design integrates generative AI and VR sketches to reinterpret cultural landmarks along the Grand Canal. *CanalWalk* combines a series of interactive mechanisms to foster cultural exploration, self-expression and social communication in visits. This research demonstrates the potential of combining generative design, spatial storytelling, and social VR to represent cultural heritage.

Index Terms: Virtual reality, exhibition, autobiographical design

1 INTRODUCTION

Cultural heritage serves as a carrier of historical memory, while simultaneously acting as a wellspring for contemporary cultural innovation. The Grand Canal, as an essential ancient transportation hub and cultural center in China, embodies abundant historical scenarios, folk cultures, and natural heritages. In this context, it is imperative not only to preserve these cultural resources, but also to explore the cultural reinterpretation of the Grand Canal through digital techniques, leading to a new form for cultural understanding, reconstruction, and re-creation.

For the past decade, there have been increasing studies that apply emerging technologies in a large number of interactive exhibitions. While current cultural dissemination mainly focuses on presentation, falling short of embodied interaction and emotional experience for audiences [1]. This hinders the development of user interests in active participation and meaning-making within cultural visits. Therefore, how to convey cultural connotations in a more immersive and interactive manner has become a critical issue. However, previous research typically focused on isolated aspects of the experience, such as digital reconstructions, visualization, gamification, generative design and personalization for interactive museum experience [2]. A more holistic approach is needed for cultural re-expression. We hope that virtual exhibitions can transcend mere 3D reproductions and passive visits that fundamentally alter the relationship between cultural institutions and audiences.

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In this paper, we curated an interactive art exhibition “*Canal-Walk*” through an autobiographical design study in VRChat, a social VR platform to explore cultural representation through digital construction and interaction techniques. The exhibition aimed to reconstruct cultural landmarks along the Grand Canal with modern artistic expression and social interaction. This research provides insights into virtual exhibition design, contributing to cultural transmission through interpretation, re-creation, and communication.

2 CANALWALK: THE VR CULTURAL EXHIBITION

We designed a virtual exhibition using an autobiographical design approach [3], which is a method drawn from ‘extensive, genuine use by those who create or build the system’. All participants were art and technology students (N=5) in our research team (M=21.40, SD=0.49). All of them have moderate experience with Generative AI (GenAI) and VR. We posited a cultural theme of ‘urban development along the canal’ in a brainstorming session and determined two key aspects to explore: 1) How to design a virtual exhibition through GenAI and immersive VR that contributes to cultural understanding, reinterpretation and transmission? 2) What design factors should be considered to improve user experiences and interactions with cultural content in a virtual exhibition?

2.1 Workflow and Design Process

We interviewed and observed participants engaged in co-creation process. Their design workflow can be concluded in three steps:

- **Background Research.** We identified representative cities as the storyline to present characteristic landscapes and rich cultural heritage along the Grand Canal through digital archives.
- **Scenes Construction.** TripoAI was used to transform landmark images from a photo-sharing website (vcg.com) into 3D models. Then, we utilized OpenBrush and Blender to refine sketches, including textures, materials, and particle effects.
- **Implementation and Testing.** We implemented the virtual exhibition in VRChat as a public world with interactive functions. To ensure cross-platform accessibility and compatibility, user tests were performed in both VR and PC settings.

Co-creation with GenAI. The authenticity and accuracy of cultural heritage are the key concerns from original images to GenAI results. We are interested in how they make efforts to bridge the gap between generated models and their visions. First, participants observed that the output often exhibited adhesion to irrelevant elements, making the subject difficult to identify. To address this issue, P1 repeatedly modified the prompts, such as “do not generate surrounding vegetation, only include main structures”, to avoid the inclusion of unrelated elements, while P4 adopted image-preprocessing to extract needed architecture. Second, they found

that the output was affected by AI hallucinations that did not align with actual architectures. For example, when P4 used prompts “*the roof tiles use glazed material*”, yet the generated textures resembled glass instead. Similar issues were observed, like incorrect architectural structures (P3) and text rendering errors (P2). When repeated generations did not yet meet expectations, all participants chose to refine the model manually. During this process, they also found some issues that increase the time cost. As P2 noted, “*to avoid system overload, these models required retopology*”. P4 further noted, “*after reducing polygons or removing irrelevant parts, the textures also need to be repainted*”.

Embodied Reconstruction in VR. To enhance cultural characteristics and digital representation of the interactive exhibition, participants use OpenBrush VR sketch tool to create stylized strokes on 3D models. During this process, participants infused their personal interpretations into the virtual exhibition with distinctive and affective expressions. In general, their design adhered to three principles: **PR1: Reconstruct entirely venues.** Cultural landmarks are not isolated from surrounding environments. Incorporating contextual visual cues into reconstruction can enhance the user presence on cultural visits. **PR2: Highlight key characteristics.** Visitors are more likely to form lasting memories at points of interest. Selecting appropriate VR brushes with visual properties that correspond to specific cultural elements and iconographies can more effectively represent cultural characteristics, thereby enhancing aesthetic and emotional resonances. **PR3: Expand cultural horizon.** Some point to the presence of religious beliefs, folk customs and local snacks as a sign of the preservation of historical culture in these cultural landmarks. Extending cultural contexts and visually representing them enable visitors to gain a deeper understanding of local culture.

2.2 Scenario Design Outcomes

Exhibition Hall A - Beijing. “*Connect the local landmark with urban development.*” The digital reconstruction of the ‘Qianhe Dew Bridge’ employed flowing light brushes to simulate long-exposure photography effects, presenting the motion trails of vehicles across the bridge (see Figure 1a). We also used glowing mesh brushes to construct the skyline and applied point light brushes to simulate the spot light shadow in urban night scenes [PR1].

Exhibition Hall B - Linqing. “*Represent classical gardening aesthetics through contemporary visual languages.*” This scenario integrates traditional culture aesthetics with futuristic elements to shape the visual design of ‘Wan Garden’ (see Figure 1b). We used particle brushes to draw water ripple effects and ambient light. In addition, glowing fog brushes were applied to produce the diffused glow of palace lanterns, creating a poetic atmosphere [PR2].

Exhibition Hall C - Jinling. “*Embody the historical and cultural heritage in ancient architecture.*” Queen Mother’s Pavilion is a well-known holy site of Taoism along the Grand Canal (see Figure 1c). This scenario aims to preserve traditional architectural features and infuse the Taoist spirit. We refer to lunar symbolism in Taoist philosophy to create a full moon behind the pavilion as the central visual [PR3]. Dynamic hydrology is realized through fluid particle strokes, and wavelike light streams reflect the historical line.

Exhibition Hall D - Yangzhou. “*Visualize the material legacy to convey cultural values.*” We chose flowing mesh brushes to outline the Grand Canal Museum (see Figure 1d). The dynamic particles generate golden mist and starburst effects that imply the value of cultural treasures inside [PR3]. This design creates an immersive space that bridges digital artistry with ancient heritage.

Exhibition Hall E - Hangzhou. “*Communicate social civilization with cultural symbols.*” Featuring an iconic lotus-petal design [PR2], the Olympic Sports Center is decorated with a gradient of blue-violet lighting effects and particle beams (see Figure 1e). The glowing mesh brushes delineate the urban silhouette of opposite bank to create visual focus [PR1]. The starry light creates a floating effect that symbolizes the sinuous water flow of the Grand Canal.

2.3 Interactive Mechanisms

Additionally, we refer to Vermeeren and Calvi’s design framework [4] to integrate mechanisms (see Figure 1) for four-stage visits.

Trigger stage: provide informative visual assets to access cultural content. *Gallery* serves as an extended area to present related images that complement primary exhibits with interactive labels. Thus, it could be a supplementary platform connecting exhibits [PR1] with detailed visual information to enhance comprehension.

Engage stage: encourage deeper exploration with regional characteristics. *Treasure-hunt* embeds collectible items related to local food culture [PR3]. Upon collecting complete sets, celebratory animations will be triggered to provide a sense of achievement. This goal-oriented exploration fosters meaningful interaction between users and cultural heritage that reinforces cultural memory.

Consolidate stage: embody personal experiences in cultural visits. *Photo-taking* allows users to take selfies or group photos with surrounding landscapes using a virtual instant camera. Therefore, photo-taking provides a way to document the user journey and reinforces the connection between virtual exhibitions and user actions. It helps users establish emotional connections with cultural content, fostering cultural communication and social sharing.

Relate stage: promote self-expressions and social interactions in creation. *Graffiti creation* provides an open space for users to draw anything using brushes. Moreover, it supports asynchronous interactions to contribute to user-generated content, fostering cultural transmission through personal and collective reinterpretation.

3 DISCUSSION AND CONCLUSION

Here, we discuss the lessons learned in this study. GenAI and VR sketches are effective mediums for co-creation to support cultural narratives and personal expressions. The process of such re-creation is not merely a digital duplication of cultural heritage, but rather a reconstruction and reinterpretation process that involves the active re-expression of cultural meanings. However, when GenAI is employed to support the design and construction of exhibitions, particular attention must be paid to how GenAI interprets and encodes cultural imagery to ensure the structural and stylistic fidelity of the generated results. This requires intervention in repetitive tasks or extra time-consuming reprocessing to ensure the rigor, authenticity, and precision of cultural representation. We also acknowledge the potential Western-centric bias in current GenAI models and thus prioritize local historical descriptors in prompting strategy to preserve the authentic visual language of Chinese culture. Additionally, our work suggests that visual effects, personalized expression, and social interaction are the main concerns to improve cultural experience in virtual exhibitions.

In conclusion, this research presents *CanalWalk*, a virtual exhibition that reconstructs cultural landscapes of cities along the Grand Canal. The exhibition design integrates generative AI, VR sketches, and digital art to create an immersive and interactive experience. *CanalWalk* offers a new pathway to foster cultural understanding and preservation through cultural reinterpretation and embodied interaction. Our study could provide valuable insights and hands-on experiences for future cultural reconstruction in digital curation.

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